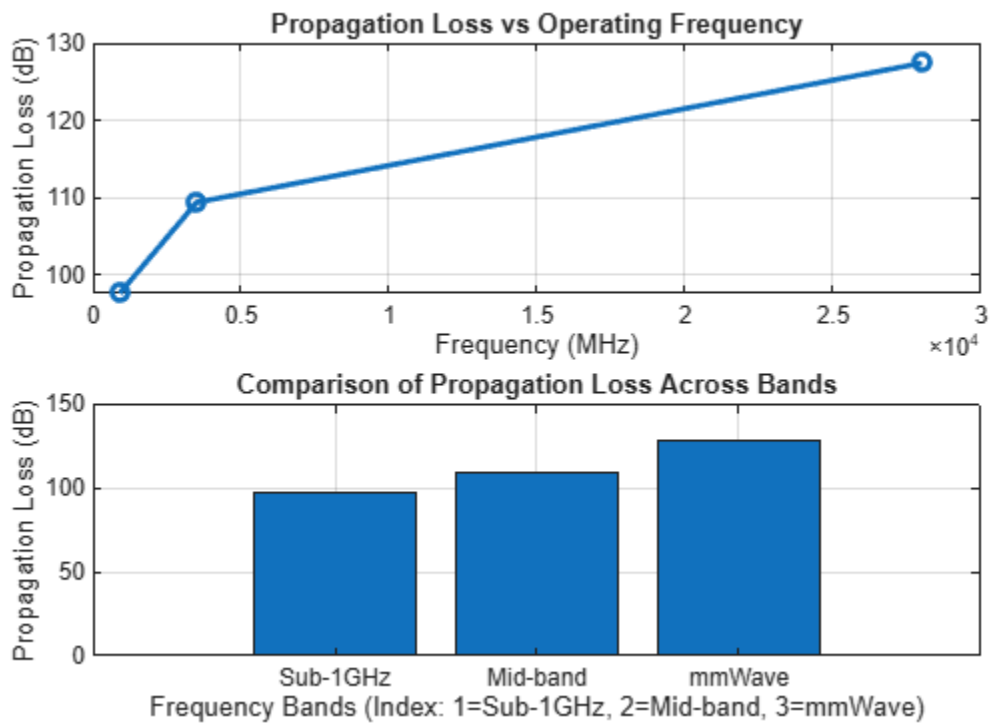


---

```

clc; clear; close all;
d = 2; % fixed distance (km)
f = [900 3500 28000]; % MHz
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% ----- Graph 1: Frequency vs Loss -----
subplot(2,1,1)
plot(f,PL, '-o', 'LineWidth',2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% ----- Graph 2: Band Comparison -----
subplot(2,1,2)
bar(PL)
title('Comparison of Propagation Loss Across Bands')
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
xticklabels({'Sub-1GHz', 'Mid-band', 'mmWave'})
grid on

```



*Published with MATLAB® R2026a*